

Geon-Woo Kim

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Research Interests

Robust and efficient ML systems and infrastructure for large-scale LLM training and serving.

Education

The University of Texas at Austin

Ph.D. in Computer Science

M.S. in Computer Science

Advisors: Aditya Akella and Daehyeok Kim

Austin, TX, USA

Nov 2026 (Expected)

Dec 2025

Seoul National University

B.S. in Computer Science & Engineering; B.S. in Mathematical Sciences (Double Major)

Seoul, South Korea

Mar 2013 - Aug 2022

- Summa Cum Laude. Completed both degrees concurrently with five years of full-time industry experience and alternative military service.

Publications

- [1] **Leto: Fast In-Place Recovery for LLM Training on Surviving Hardware**, *Geon-Woo Kim*, Joon Ha Kim, Daehyeok Kim, arXiv preprint arXiv:2610.00687 (**Preprint 2026**)
- [2] **StreamEP: Straggler-Tolerant MoE Decoding without Communication Barriers**, Yizhuo Liang, Shaoyu Wang, Jaeyong Song, Yanqi Zhou, *Geon-Woo Kim*, Guangrong He, Seo Jin Park, ACM Symposium on Operating Systems Principles (**SOSP 2026**)
- [3] **Dooly: Configuration-Agnostic, Redundancy-Aware Profiling for LLM Inference Simulation**, Joon Ha Kim, *Geon-Woo Kim*, Anoop Rachakonda, Daehyeok Kim, Fortieth Annual Conference on Neural Information Processing Systems (**NeurIPS 2026**)
- [4] **Reforge: Low-Latency Distributed GNN Serving with Selective Embedding Recomputation**, *Geon-Woo Kim*, Donghyun Kim, Jeongyoon Moon, Henry Liu, Tarannum Khan, Anand Iyer, Daehyeok Kim, Aditya Akella, IEEE International Parallel and Distributed Processing Symposium (**IPDPS 2026**)
- [5] **HALoS: Hierarchical Asynchronous Local SGD over Slow Networks for Geo-Distributed Large Language Model Training**, *Geon-Woo Kim*, Junbo Li, Shashidhar Gandham, Omar Baldonado, Adithya Gangidi, Pavan Balaji, Zhangyang Wang, Aditya Akella, Forty-second International Conference on Machine Learning (**ICML 2025**)
- [6] **StitchLLM: Serving LLMs, One Block at a Time**, Bodun Hu, Shuoze Li, Saurabh Agarwal, Myungjin Lee, Akshay Jajoo, Jiamin Li, Le Xu, *Geon-Woo Kim*, Donghyun Kim, Hong Xu, Amy Zhang, Aditya Akella, The 63rd Annual Meeting of the Association for Computational Linguistics (**ACL 2025**)
- [7] **Read-ME: Refactorizing LLMs as Router-Decoupled Mixture of Experts with System Co-Design**, Ruisi Cai, Yeonju Ro, *Geon-Woo Kim*, Peihao Wang, Babak Ehteshami Bejnordi, Aditya Akella, and Zhangyang Wang, Thirty-Eighth Annual Conference on Neural Information Processing Systems (**NeurIPS 2024**)
- [8] **Lovelock: Towards Smart NIC-hosted Clusters**, Seo Jin Park, Ramesh Govindan, Kai Shen, David Culler, Fatma Özcan, *Geon-Woo Kim*, and Hank Levy, HotCarbon Workshop on Sustainable Computer Systems (**HotCarbon 2024**)
- [9] **Orca: A distributed serving system for Transformer-Based generative models**, Gyeong-In Yu, Joo Seong Jeong, *Geon-Woo Kim*, Soojeong Kim, and Byung-Gon Chun, USENIX Symposium on Operating Systems Design and Implementation (**OSDI 2022**)
- [10] **Terra: Imperative-Symbolic Co-Execution of Imperative Deep Learning Programs**, Taebum Kim, Eunji Jeong, *Geon-Woo Kim*, Yunmo Koo, Sehoon Kim, Gyeong-In Yu, and Byung-Gon Chun, Thirty-Fifth Annual Conference on Neural Information Processing Systems (**NeurIPS 2021**)
- [11] **Apache Nemo: A Framework for Building Distributed Dataflow Optimization Policies**, Youngseok Yang, Jeongyoon Eo, *Geon-Woo Kim*, Joo Yeon Kim, Sanha Lee, Jangho Seo, Won Wook Song, and Byung-Gon Chun, USENIX Annual Technical Conference (**ATC 2019**)
- [12] **Pado: A data processing engine for harnessing transient resources in datacenters**, Youngseok Yang, *Geon-Woo Kim*, Won Wook Song, Yunseong Lee, Andrew Chung, Zhengping Qian, Brian Cho, and Byung-Gon Chun, ACM European Conference on Computer Systems (**EuroSys 2017**)

Research Experience

Graduate Research Assistant, UTNS

UT Austin

Advisors: Aditya Akella and Daehyeok Kim

Sep 2022 - Current

- *Leto (Preprint 26)*: Built a fault-tolerant LLM training system that recovers in place from *hardware-operable failures*, which leave GPUs usable without repair, by retaining model and process state on surviving GPUs instead of reloading checkpoints; on A100 clusters, recovers **3.6–6.5× faster** than the best checkpointing baselines and adds **up to 13.7 percentage points** of productive training time; simulation shows **over 95%** productive time at 131K-GPU scale.
- *HALoS (ICML 25)*: Built a hierarchical asynchronous optimization framework for geo-distributed LLM training in collaboration with Meta; converges **up to 7.5× faster** than synchronous training and **2.1× faster** than prior asynchronous methods while matching model quality, with formal convergence guarantees.
- *Reforge (IPDPS 26)*: Built a distributed GNN serving system that cuts serving latency by **up to 159×** over full-graph serving and **10.8×** over approximation baselines with <1% accuracy loss, via selective embedding recomputation and computation-graph parallelism.

Student Researcher, SystemsResearch@Google

Google

Mentor: Seo Jin Park

May 2023 - Jul 2023

- *StreamEP (SOSP 26)*: Proposed and simulated fine-grained expert disaggregation for MoE LLM serving; the project continued with collaborators and culminated in a barrier-free expert-parallel decoding system that achieves **up to 1.7×** higher throughput than SGLang on commodity GPU clusters.

Undergraduate Research Intern, SPL

SNU

Advisor: Byung-Gon Chun

Jan 2015 - Aug 2021

- *Orca (OSDI 22)*: Core contributor to the serving system that introduced **continuous batching**, now the standard technique adopted in vLLM, TensorRT-LLM, and SGLang; extended attention CUDA kernels to enable fine-grained KV cache management; and built the end-to-end serving path, from the low-latency C++ web frontend to distributed multi-GPU deployment.
- Integrated NVIDIA CUPTI into Orca's C++ engine and enabled browser-based GPU kernel-trace visualization for performance profiling and debugging.
- *Terra (NeurIPS 21)*: Enabled symbolic execution of imperative TensorFlow programs; strengthened formal guarantees with JIT compilation and mathematical proof, and extensively profiled and optimized the system.

Work Experience

Software Engineer Intern, Machine Learning (PhD)

Meta Platforms, Inc.

Mentor: Saif Hasan (AI Networking Team)

May 2025 - Aug 2025

- Developed **NVIDIA-AMD heterogeneous GPU collectives** (AllReduce, AllToAll) in NCCLX, Meta's collective communication library (open-sourced as [torchcomms](#)), across RDMA NICs and NVLink/xGMI interconnects.
- Made **100%** of the NCCLX codebase compilable on both CUDA and ROCm with single-source builds.
- Built an RDMA performance tool for mixed-vendor environments (NVIDIA-AMD GPUs, Mellanox-Broadcom NICs).

Software Platform Engineer

Viva Republica (Toss)

Korea's leading fintech, valued at \$7B+; joined as the 3rd server developer

Jun 2016 - Feb 2021

- Proposed and led TUBA, Toss's company-wide mobile marketing platform for A/B testing, large-scale targeted messaging, and real-time, log-driven in-app prompts, serving 20M+ users.
- Built and maintained a large-scale data analysis platform on Apache Spark, HDFS, Impala, Flink, and PyTorch.

Professional Service

2026 **Reviewer (NeurIPS)** Advances in Neural Information Processing Systems

2026 **Reviewer (TPDS)** IEEE Transactions on Parallel and Distributed Systems

2026 **Artifact Evaluation Committee (OSDI)** Symposium on Operating Systems Design and Implementation

Honors & Awards

Aug 2022 **Overseas PhD Scholarship**

Korea Foundation for Advanced Studies

Mar 2013 **Presidential Science Scholarship**

Korean Government